Unit 1: Number System

Teaching Point: Start the lesson by explaining what a "number system" is—a mathematical way to represent numbers using a specific set of digits or symbols. Introduce the crucial concept of a **"Base"** or **"Radix,"** which simply means the total number of unique digits or symbols used in that specific system.

1. Decimal Number System

- **Teaching Point:** Relate this to what students already use every day. It is the standard human counting system.
- **Key Concepts:**
  - **Base:** 10
  - **Digits Used:** 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
  - *How it works:* It is a positional system based on powers of 10. For example, in the number 345, the 5 is in the ones place (), 4 is in the tens place (), and 3 is in the hundreds place ().

2. Binary Number System

- **Teaching Point:** Emphasize that this is the absolute foundation of computing. The computer's hardware (CPU and memory) only understands this system.
- **Key Concepts:**
  - **Base:** 2
  - **Digits Used:** 0 and 1
  - *Analogy:* Explain that 0 and 1 represent physical electrical states inside the computer's transistors: 0 means "Off" (low voltage) and 1 means "On" (high voltage).

3. Octal Number System

- **Teaching Point:** Explain this as a system historically used to simplify binary numbers. Because 8 is a power of 2 (), one octal digit can perfectly represent exactly three binary digits.
- **Key Concepts:**
  - **Base:** 8
  - **Digits Used:** 0, 1, 2, 3, 4, 5, 6, 7 (Notice there is no 8 or 9).

- *Context:* While less common today, it was popular in early computing and is still used in systems like Unix/Linux for setting file permissions.

4. Hexadecimal Number System

- **Teaching Point:** Describe this as a highly compact system used heavily in modern computing. Because 16 is a power of 2 (), one hexadecimal digit represents exactly four binary digits (a "nibble").
- **Key Concepts:**
  - **Base:** 16
  - **Digits/Symbols Used:** 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, **A, B, C, D, E, F**.
  - *Explanation:* Since we run out of single digits after 9, we use letters where A=10, B=11, C=12, D=13, E=14, and F=15.
  - *Real-World Examples for Students:* Hexadecimal is what they see in Wi-Fi MAC addresses, IPv6 addresses, error codes (like 0x80004005), and web color codes (e.g., #FF0000 for pure red).

5. Number Conversion

- **Teaching Point:** This is the mathematical core of the unit. Teach students the step-by-step algorithms to translate numbers between different bases.
- **Key Conversions to Practice on the Board:**
  - **Decimal to Binary/Octal/Hex:** Teach the **"Successive Division"** method. Divide the decimal number by the target base (e.g., divide by 2 to get binary) and record the remainders. Read the remainders from bottom to top to get the answer.
  - **Binary/Octal/Hex to Decimal:** Teach the **"Positional Weight"** method. Multiply each digit by its base raised to the power of its position index (starting from 0 on the far right) and sum the results.
  - **Binary to Hexadecimal (and vice versa):** Teach the **"Grouping"** method. To convert binary to hex, group the binary string into chunks of four (starting from the right) and translate each 4-bit chunk into its single hexadecimal equivalent.